Relay — AI Workflow Orchestrator
Node.js · TypeScript · NestJS · PostgreSQL

Goal: A durable workflow engine with exactly-once side effects, schema-validated AI decisions, and approval gates the model cannot bypass.

1. **Architecture**
   - The engine core has no framework dependency and sits behind ports (Repository, Queue, AIProvider, WorldClient). NestJS is only the API layer.
   - The API and worker are separate processes. The API only enqueues work.
2. **Workflows & Triggers**
   - Draft → publish. Published workflows are immutable. Every run executes its own snapshot of the definition.
   - Publish validation against the catalog rejects unknown types, missing params and dangling references. Loops are allowed.
   - Webhooks are secured by X-Relay-Secret (constant-time compare). A manual trigger is also provided.
3. **Execution**
   - queued → running ⇄ waiting_approval → succeeded | failed | cancelled
   - Each step persists its input, output, attempts, timing and tokens.
   - Delays and retries are re-queued. A worker never sleeps holding a job.
4. **Durability & Exactly-Once**
   - Postgres is both the store and the queue (SKIP LOCKED plus a lease). Expired jobs are reclaimed.
   - Before a side effect: save the step with key {run_id}:{node_id}:{sequence}. Then call. Then save the result.
   - On resume, the same key is sent again, so the mock world ignores the duplicate.
   - Every external call has a timeout. Retries use backoff, and 4xx errors fail fast.
5. **AI & Guardrails**
   - Ajv validates AI output against output_schema, with one repair retry before the step fails.
   - Payloads are treated as data. AI output only selects a branch; it never chooses targets or actions.
   - Only the approve API can approve a run. The engine checks for that approval before any sensitive node.
   - The step cap is checked before every node; wf_runaway stops at 12.
6. **Testing & Verification**
   - Tests: publish validation, templates, schema enforcement, idempotency, gate bypass, injection payloads, step cap.
   - Drills: smoke test, then kill -9 of the worker followed by duplication_check.py.
7. **Deliverables**
   - Console, README, verification report, demo video. Stretch goal: NL compiler.

Success criteria: smoke test passes · no duplicate side effects after a crash · invalid AI output rejected · injections never bypass approval · runaway loop stopped.